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Imaging device

Abstract

Provided is a technique for detecting deviation amounts in two directions between a tip of a nozzle of a pipetting device and a target stop position from one image. An imaging device of the present disclosure is an imaging device installed in a pipetting device including a nozzle that aspirates and dispenses liquid and an arm that holds the nozzle and moves the nozzle by a rotation operation. The imaging device includes: a camera that includes an image sensor and a lens; and a pair of mirrors that includes a first mirror having a first reflective surface and a second mirror having a second reflective surface. The pair of mirrors is arranged such that the first reflective surface and the second reflective surface face each other. The pair of mirrors is arranged one by one between the tip of the nozzle and a bottom surface of the arm, on both sides across a plane that includes a rotation axis of the arm and a central axis of the nozzle. The first mirror is disposed closer to the arm than the second mirror, with the first reflective surface facing toward the tip of the nozzle, and the second reflective surface facing toward the camera.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6359694	12/2001	Stredele et al.	N/A	N/A
6435808	12/2001	Araya et al.	N/A	N/A
2018/0275053	12/2017	Obara	N/A	G01N 1/38
2018/0313773	12/2017	Broga	N/A	B01L 3/52
2019/0339189	12/2018	Takeda	N/A	G01N 37/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2872765	12/1998	JP	N/A
2005-049197	12/2004	JP	N/A
2012-032310	12/2011	JP	N/A
2015-172509	12/2014	JP	N/A
2017-151002	12/2016	JP	N/A

OTHER PUBLICATIONS

International Search Report with English Translation and Written Opinion of International Patent Application No. PCT/JP2022/019476 dated Jun. 14, 2022 (9 pages). cited by applicant
Extended European Search Report issued in corresponding EP Patent Application No. 22822848.2, dated Apr. 14, 2025 (9 pages). cited by applicant

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to an imaging device.

BACKGROUND ART

(2) A nozzle used for pipetting a specimen or a reagent in an automatic analyzer is periodically replaced from the viewpoint of maintaining the pipetting accuracy. At the time of pipetting, it is necessary to insert the tip of the nozzle into a plurality of narrow spaces, such as a specimen container, a reagent container, or a cleaning hole. Therefore, each time the nozzle is replaced, the position of the tip of the nozzle needs to be adjusted with respect to the target stop position, such as the specimen container, the reagent container, or the cleaning hole.

(3) In PTL 1, the problem to be solved is to “provide a specimen processing device capable of adjusting the position of a moving part more easily and precisely than conventional devices, regardless of the complexity of the moving path of the moving part”. In PTL 1, the following technique is disclosed: “When an abnormality occurs in a specimen analyzer, a camera 23d provided in the specimen analyzer captures an image of an object (cuvette) used for adjusting a mechanism (reagent pipetting unit 23) where the abnormality has occurred. The adjustment amount for the reagent pipetting unit 23 is detected based on the image obtained by imaging, and the reagent pipetting unit 23 is adjusted by the detected adjustment amount.” (cf. Abstract of PTL 1).

(4) In PTL 2, the problem to be solved is to “obtain a nozzle tip position measuring device in which damage to a nozzle tip or contamination of foreign matter does not occur, and which can accurately measure precise position information in the micrometer order”. In PTL 2, the following technique is disclosed: “The device measures the tip position of a nozzle, which is provided on a moving means capable of relative movement in the XYZ triaxial directions, each orthogonal to the XY plane, on a base. The tip of the nozzle is equipped with a reference area imaging means that captures an image in a reference area determined beforehand on the base. A position displacement from the reference position of the tip of the nozzle, determined beforehand in the reference area, is measured using an image captured when the tip of the nozzle is moved to the reference position.” (cf. Abstract of PTL 2).

CITATION LIST

Patent Literature

(5) PTL 1: JP 2012-32310

(6) PTL 2: JP 2005-49197

SUMMARY OF INVENTION

Technical Problem

(7) In the technique described in PTL 1, an image of the tip of a pipette is captured at a target stop position, and detected is a deviation in the left-right direction between the center line of the acquired image and the target stop position as the adjustment amount. However, it is impossible to detect the deviation amount in the depth direction of the image. Thus, even when both the tip of the pipette and the target stop position coincide with the central axis of the image on the image, the tip of the nozzle may be adjusted to a position that deviates in the depth direction from the target stop position.

(8) In the technique described in PTL 2, the reference position for adjusting the tip position of the nozzle is not the target stop position itself. Accordingly, when there is a deviation on the side of the

target stop position, such as the specimen container, the reagent container, or the cleaning hole, it may not be possible to adjust the tip of the nozzle to the target stop position.

(9) Therefore, the present disclosure provides a technique capable of detecting deviation amounts in two different directions between the tip of a nozzle of a pipetting device and a target stop position from one captured image.

Solution to Problem

(10) In order to solve the above problem, an imaging device according to the present disclosure is an imaging device installed in a pipetting device including a nozzle configured to aspirate and dispense liquid and an arm configured to hold the nozzle and to move the nozzle by a rotation operation. The imaging device includes: a camera that includes an image sensor and a lens; and a pair of mirrors that includes a first mirror having a first reflective surface and a second mirror having a second reflective surface. The pair of mirrors is arranged such that the first reflective surface and the second reflective surface face each other. The pair of mirrors is arranged one by one between a tip of the nozzle and a bottom surface of the arm, on both sides across a plane that includes a rotation axis of the arm and a central axis of the nozzle. The first mirror is disposed closer to the arm than the second mirror, with the first reflective surface facing toward the tip of the nozzle, and the second reflective surface facing toward the camera.

(11) Further features related to the present disclosure will become apparent from the description of the present specification and the accompanying drawings. The aspects of the present disclosure are achieved and realized by elements, combinations of various elements, the following detailed description, and aspects of the appended claims. The description of the present specification is merely exemplary, and does not limit the scope of claims or application examples of the present disclosure in any sense.

Advantageous Effects of Invention

(12) According to the technology of the present disclosure, it is possible to detect deviation amounts in two different directions between the tip of the nozzle of the pipetting device and the target stop position from one captured image. Problems, configurations, and effects other than those described above will be clarified by the following description of embodiments.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic configuration diagram of an automatic analyzer.

(2) FIG. 2 is a side view of a pipetting device with an imaging device attached thereto.

(3) FIG. 3 is a rear view of the periphery of the imaging device of the pipetting device with the imaging device attached thereto.

(4) FIG. 4A is a diagram for explaining an optical path of object light incident on the image sensor.

(5) FIG. 4B is a diagram for explaining an optical path of object light incident on the image sensor.

(6) FIG. 5 is a schematic diagram illustrating an example of an image of a nozzle captured by the imaging device.

(7) FIG. 6 is a rear view of the periphery of the imaging device of the pipetting device with the imaging device attached thereto.

(8) FIG. 7 is a bottom view of the periphery of the imaging device of the pipetting device with the imaging device attached thereto.

(9) FIG. 8A is a diagram for explaining an optical path of object light incident on the image sensor.

(10) FIG. 8B is a diagram for explaining an optical path of object light incident on the image sensor.

(11) FIG. 9 is a schematic diagram illustrating an example of an image of a nozzle captured by the imaging device.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(12) <Configuration Example of Automatic Analyzer>

(13) FIG. 1 is a schematic configuration diagram of an automatic analyzer **10**. The automatic analyzer **10** includes a reagent disk **12**, a reaction disk **13**, a plurality of reagent pipetting devices **14**, a plurality of specimen pipetting devices **15**, a conveyance line **16**, a cleaning tank **17**, and a control device **100**.

(14) The reagent disk **12** holds a plurality of reagent containers **11**. The reaction disk **13** is configured to be rotatable and holds a plurality of reaction cells **20** arranged in the circumferential direction. The conveyance line **16** conveys a rack **18**. The rack **18** holds a plurality of specimen containers **19**. The reagent pipetting device **14** includes a nozzle **141** for pipetting a reagent (liquid) contained in the reagent container **11**. The reagent pipetting device **14** is configured to be capable of moving the nozzle **141** in the horizontal direction and the vertical direction. The reagent pipetting device **14** aspirates the reagent into the nozzle **141** and dispenses the reagent to the reaction cell **20**. The specimen pipetting device **15** includes the nozzle **151** for pipetting a specimen (liquid) contained in the specimen container **19**. The specimen pipetting device **15** is configured to be capable of moving the nozzle **151** in the horizontal direction and the vertical direction. The specimen pipetting device **15** aspirates the specimen into the nozzle **151** and dispenses the specimen to the reaction cell **20**. The specimen is, for example, a biological sample derived from blood, such as serum or whole blood, or derived from urine. The reagent and the specimen pipetted into the reaction cell **20** are stirred by a stirring device (not illustrated) to obtain a reaction liquid. The cleaning tank **17** cleans the nozzle **151** of the specimen pipetting device **15**.

(15) The reagent pipetting device **14** moves the nozzle **141** to a stop position for aspirating the reagent from the reagent container **11**, a stop position for discharging the reagent into the reaction cell **20**, and a stop position for washing off the reagent attached to the nozzle **141** in a cleaning tank (not illustrated). Similarly, the specimen pipetting device **15** moves the nozzle **151** to a stop position for aspirating the specimen from the specimen container **19**, a stop position for discharging the specimen to the reaction cell **20**, and a stop position for washing off the specimen attached to the tip of the nozzle **151** in the cleaning tank **17**. The reagent pipetting device **14** moves up and down the nozzle **141** in accordance with the height of each stop position. The specimen pipetting device **15** moves up and down the nozzle **151** in accordance with the height of each stop position.

(16) The control device **100** can be formed of a computer device. The control device **100** includes a processor **101**, a storage device **102**, an input device **103**, and an output device **104**. The processor **101** controls the operation of the automatic analyzer **10** according to a program stored in the storage device **102**. The processor **101** analyzes the reaction liquid in the reaction cell **20**. The storage device **102** can be formed of, for example, an internal memory or an external storage, and stores programs and parameters necessary for processing of the processor **101**. The input device **103** can be formed of, for example, a mouse, a keyboard, a touch panel, or the like. The output device **104** can be formed of, for example, a display, a speaker, a touch panel, or the like.

(17) <Configuration Example of Pipetting Device>

(18) FIG. 2 is a side view of the reagent pipetting device **14** with an imaging device **200** attached thereto. The reagent pipetting device **14** includes the nozzle **141**, arms **142**, **143**, and a shaft **144**. The base end portion of the nozzle **141** is held by one end portion of the arm **142**. The arm **142** is rotatably coupled to the arm **143**. FIG. 2 illustrates a state where the arm **142** and the arm **143** are arranged in a straight line. The arm **143** is supported by the shaft **144**. The shaft **144** is configured to be rotatable and vertically movable by a motor (not illustrated). Note that the arm **143** may not be provided, and in this case, the arm **142** is connected to the shaft **144**.

(19) As illustrated in FIG. 2, in the present specification, the longitudinal direction of the shaft **144** is defined as the vertical direction. A direction parallel to the longitudinal direction of the arm **142** is defined as the front-back direction, with the front denoted by F and the back denoted by B. An

optical axis of a camera **220**, that is, a straight line passing through the center of an image sensor **221** and the principal point of a lens **222**, is denoted by O.

(20) The imaging device **200** includes a pair of mirrors **210** and the camera **220**. The pair of mirrors **210** includes a first mirror **211** and a second mirror **212** arranged such that the reflective surfaces face each other. That the reflective surfaces “face each other” does not mean that the reflective surface of the first mirror **211** and the reflective surface of the second mirror **212** face each other in parallel, but means that light reflected from the reflective surface of the first mirror **211** is incident on the reflective surface of the second mirror **212**. The camera **220** is a monocular camera including an image sensor **221** and a lens **222**. The imaging device **200** is attached to the arm **142** by an attachment member (not illustrated). The camera **220** is disposed below the bottom surface of the arm **142** and behind the nozzle **141**. The first mirror **211** and the second mirror **212** are disposed between the tip of the nozzle **141** and the bottom surface of the arm **142**. The attachment member includes support members that support each of the first mirror **211** and the second mirror **212**, respectively.

(21) FIG. **3** is a rear view of the periphery of the imaging device **200** as viewed from the back. As illustrated in FIG. **3**, in the present specification, the lateral direction (a direction perpendicular to the vertical direction and the front-back direction) of the arm **142** is defined as the left-right direction, with the right denoted by R and the left denoted by L. A plane including a rotation axis Q of the arm **142** and the central axis of the nozzle **141** is denoted by A. A plane including the center of the image sensor **221** and the central axis of the nozzle **141** is denoted by P. In the present embodiment, the plane A and the plane P are the same.

(22) The first mirror **211** is disposed on a right R of the plane A. The second mirror **212** is disposed on a left L of the plane A. That is, the first mirror **211** and the second mirror **212** are disposed one by one on both sides, the left L and the right R, across the plane A. The reflective surface of the first mirror **211** faces toward the tip of the nozzle **141**. The reflective surface of the second mirror **212** faces toward the base end portion of the nozzle **141**. The first mirror **211** is disposed closer to the base end portion of the nozzle **141** than the second mirror is, that is, disposed above the second mirror **212**.

(23) Since the pair of mirrors **210** is arranged as described above, the images of the tip portion of the nozzle **141** viewed from viewpoints in two different directions are formed on the image sensor **221**, which will be described later in detail. The image sensor **221** outputs the captured images to the control device **100**. The control device **100** processes the images captured by the image sensor **221** and calculates deviation amounts in two directions between the tip of the nozzle **141** and the target stop position. The control device **100** calculates the adjustment amount of the position of the nozzle **141** based on the calculated deviation amounts, and adjusts the operation amount of the motor of the reagent pipetting device **14** based on the adjustment amount.

(24) <Captured Image>

(25) FIGS. **4A** and **4B** are diagrams for explaining optical paths of object light incident on the image sensor **221**. FIG. **4A** illustrates an optical path **51** of the object light reflected by the front side surface of the tip portion of the nozzle **141**. FIG. **4B** illustrates an optical path **52** of the object light reflected by the right side surface of the tip portion of the nozzle **141**. As indicated by the optical path **51** in FIG. **4A**, the object light reflected by the front side surface of the tip portion of the nozzle **141** is directly incident on the lens **222** and forms an image in a first region substantially at the center of the image sensor **221**. On the other hand, as indicated by the optical path **52** in FIG. **4B**, the object light reflected by the right side surface of the tip portion of the nozzle **141** is once reflected by the first mirror **211** toward the tip of the nozzle **141**, then reflected by the second mirror **212** toward the camera **220**, and forms an image in a second region at the right end portion of the image sensor **221**. Therefore, in the present embodiment, images of the tip portion of the nozzle **141** viewed from two different viewpoints are displayed on one image sensor **221**.

(26) An example of the stop position of the nozzle **141** is a cleaning hole disposed at the bottom of

the cleaning tank **17**. The nozzle **141** is inserted into the cleaning hole and then cleaned, thus requiring the adjustment of the position of the nozzle **141** to the center of the cleaning hole.

(27) FIG. 5 is a schematic diagram illustrating an example of an image of the nozzle **141** stopped above a cleaning hole **171** of the cleaning tank **17** captured by the imaging device **200**. As illustrated in FIG. 5, in one image, an image **221F** is captured in a first imaging region and an image **221R** is captured in a second imaging region. The image **221F** is from a viewpoint placed on the front side surface of the tip portion of the nozzle **141**. The image **221R** is from a viewpoint placed on the right side surface of the tip portion of the nozzle **141**. Therefore, it is possible to detect the deviation amount in the left-right direction between the center of the cleaning hole **171** (the target stop position) and the tip of the nozzle **141** from the image **221F** as a distance DH between the intersection of a central axis **141FC** of the nozzle **141** and a tip edge **141FE** of the nozzle **141** and a center line **171FC** in the left-right direction of the cleaning hole **171**. Similarly, it is possible to detect the deviation amount in the front-back direction between the center of the cleaning hole **171** and the tip of the nozzle **141** from the image **221R** as a distance DV between the intersection of a central axis **141RC** of the nozzle **141** and a tip edge **141RE** of the nozzle **141** and a center line **171RC** in the vertical direction of the cleaning hole **171**.

(28) As described above, according to the imaging device **200** of the present embodiment, with the pair of mirrors **210** being arranged as described above, the images of the tip portion of the nozzle **141** viewed from two different viewpoints are displayed on one image sensor **221**. Accordingly, it is possible to detect, from one captured image, in two directions of the left-right direction and the front-back direction between the tip position of the nozzle **141** and the target stop position. In this manner, a deviation in the left-right direction is detected from one of the two regions included in one captured image, and a deviation in the front-back direction is detected from the other region. Thereby, it is possible to reduce the number of pixels to be processed at a time compared to the case of processing all the regions in one captured image, resulting in an improvement in processing speed.

(29) Note that the directions of the viewpoints, from which imaging is performed, are not limited to the left-right direction and the front-back direction, but may be two intersecting directions. In particular, imaging from viewpoints in two orthogonal direction enables precise detection of a deviation between the tip of the nozzle **141** and the target stop position. Note that the term “orthogonal” in the present specification includes not only a mode in which two directions intersect strictly at an angle of 90° but also a mode in which two directions intersect at an angle within a range of $90^\circ \pm \text{error}$ (substantially orthogonal).

(30) As described above, in the first embodiment, it has been described that the imaging device **200** is attached to the reagent pipetting device **14**. The imaging device **200** can be similarly attached to the specimen pipetting device **15**. In this case, the imaging device **200** can detect deviations in two directions from one captured image by capturing images of the position of the nozzle **151** of the specimen pipetting device **15**.

(31) It has been described that the imaging device **200** is attached to the pipetting device of the automatic analyzer **10** has been described. The imaging device **200** is also applicable to pipetting devices mounted on other devices.

(32) <Modification of First Embodiment>

(33) The imaging device **200** may be configured to be detachable from the pipetting device. The imaging device **200** may be attached to the pipetting device only during the position adjustment for the nozzle and detached during normal analysis operation. In this case, there is no mass of the imaging device **200** during the analysis operation, thereby the load on the motor that drives the pipetting device can be reduced. As a result, the throughput of the analysis can be improved.

(34) The second mirror **212** may be disposed closer to the optical axis O of the camera **220** than the first mirror **211**. In this case, on the captured image, the area of the second imaging region, where an image is captured via the first mirror **211** and the second mirror **212**, increases. This is

advantageous for recognition of the positional relationship in the front-back direction between the tip of the nozzle **141** and the target stop position. In this case, the position of the nozzle **141** on the captured image is the same as the position illustrated in FIG. 5.

(35) When a single focal length lens is used for the lens **222**, the second mirror **212**, which is disposed on the tip side of the nozzle **141**, can be made smaller than the first mirror **211**. This is because the second mirror **212** is disposed at a position closer to the camera **220** than the first mirror **211** on the optical path **52** where object light from the right side surface of the tip portion of the nozzle **141** reaches the second imaging region via the first mirror **211** and the second mirror **212**. This enables the nozzle **141** to move between the plurality of target stop positions without a collision of the imaging device **200** with other components in the automatic analyzer **10**.

(36) Glass (refractive index>1) having a longitudinal direction in the optical axis direction of the camera **220** may be disposed on the optical path **51** and outside the optical path **52**. In this case, the focus of the camera **220** is set at a position separated from the image sensor **221** by the optical path length of the optical path **52**. This can equalize the optical path length of the image **221F** and the optical path length of the image **221R**, thus improving image resolution.

(37) <Summary of First Embodiment>

(38) As described above, the imaging device **200** according to the first embodiment is installed in the reagent pipetting device **14** including the nozzle **141** that aspirates and dispenses liquid and the arm **142** that holds the nozzle **141** and moves the nozzle **141** by a rotational operation. The imaging device **200** includes the camera **220** including the image sensor **221** and the lens **222**, and the pair of mirrors **210** including the first mirror **211** and the second mirror **212**. The pair of mirrors **210** is arranged such that the reflective surface of the first mirror **211** faces the reflective surface of the second mirror **212**. The pair of mirrors **210** is arranged one by one between the tip of the nozzle **141** and the bottom surface of the arm **142**, on both sides across the plane A that includes the rotation axis Q of the arm **142** and the central axis of the nozzle **141**. The first mirror **211** is disposed closer to the arm **142** than the second mirror **212**. The reflective surface of the first mirror **211** faces toward the tip of the nozzle **141**. The reflective surface of the second mirror **212** faces toward the camera **220**.

(39) According to these configurations, an image of the tip portion of the nozzle **141** viewed from a first viewpoint is formed in a first imaging region of the image sensor **221**. An image of the tip portion of the nozzle **141** viewed from a second viewpoint is formed in a second imaging region of the image sensor **221**. Therefore, one captured image includes images of the nozzle **141** viewed from two different viewpoints. Accordingly, it is possible to detect deviation amounts in two different directions between the tip of the nozzle **141** of the reagent pipetting device **14** and the target stop position.

Second Embodiment

(40) In the second embodiment, another example of the positional relationship between the camera **220** and the pair of mirrors **210** will be described. In the drawings, elements having the same functions as those of the first embodiment are denoted by the same reference numerals, and the description thereof will be omitted.

(41) FIG. 6 is a rear view of the periphery of the imaging device **200** according to a second embodiment as viewed from the back. In the first embodiment described above, it has been described that the optical axis O of the camera **220** and the central axis of the nozzle **141** are arranged on the same plane A. In contrast, in the present embodiment, as illustrated in FIG. 6, the camera **220** is disposed to be translated to the left L side such that the optical axis O of the camera **220** is located on the left L side with respect to the plane A defined in the first embodiment.

(42) FIG. 7 is a bottom view of the periphery of the imaging device **200** of the reagent pipetting device **14** according to the second embodiment. As illustrated in FIG. 7, the optical axis O of the camera **220** is disposed outside the plane P including the center of the image sensor **221** and the central axis of the nozzle.

(43) FIGS. 8A and 8B are diagrams for explaining optical paths of object light incident on the image sensor 221. FIG. 8A illustrates an optical path 53 of the object light reflected by the front surface of the tip portion of the nozzle 141. FIG. 8B illustrates an optical path 54 of the object light reflected by the right side surface of the tip portion of the nozzle 141. As illustrated by the optical path 53 in FIG. 8A, the object light reflected by the front side surface of the tip portion of the nozzle 141 is directly incident on the lens 222 and forms an image in a first region on the left half of the image sensor 221. On the other hand, as illustrated by the optical path 54 in FIG. 8B, the object light reflected by the right side surface of the tip portion of the nozzle 141 is once reflected by the first mirror 211 toward the tip of the nozzle 141, then reflected by the second mirror 212 toward the camera 220, and forms an image in a second region on the right half of the image sensor 221. Therefore, in the present embodiment, images of the tip portion of the nozzle 141 from different viewpoints are projected with substantially the same area on one image sensor 221.

(44) FIG. 9 is a schematic diagram illustrating an example of an image of the nozzle 141 stopped above the cleaning hole 171 of the cleaning tank 17 captured by the imaging device 200. As illustrated in FIG. 9, in the present embodiment, an area of an image 321F in the first imaging region and an area of an image 321R in the second imaging region are substantially equal. The tip portion of the nozzle 141 can be projected near the center of each of the first imaging region and the second imaging region. Therefore, there is no unnecessary region on the captured image. It is possible to more accurately detect deviation amounts in two directions of the left-right direction and the front-back direction between the tip of the nozzle 141 and the target stop position. In this manner, according to the imaging device 200 of the present embodiment, it is possible to more efficiently detect deviations in two directions of the left-right direction and the front-back direction between the tip position of the nozzle 141 and the target stop position from one captured image.

(45) <Summary of Second Embodiment>

(46) As described above, in the imaging device 200 according to the second embodiment, the optical axis O of the camera 220 is disposed outside the plane A including the rotation axis Q of the arm 142 and the central axis of the nozzle. As a result, the area of the image 321F in the first imaging region of the image sensor 221 and the area of the image 321R in the second imaging region can be made substantially equal. Consequently, it is possible to detect more accurately deviation amounts of the tip of the nozzle 141 in two directions.

Modification

(47) The present disclosure is not limited to the above embodiments, and includes various modifications. For example, the above embodiments have been described in detail to describe the present disclosure in an easy-to-understand manner, and do not necessarily have all the configurations described. It is possible to replace a part of one embodiment with components of another embodiment. It is also possible to add components of another embodiment to the components of one embodiment. Furthermore, it is possible to perform addition, deletion, or replacement for some of the components of each embodiment, using some of the components of another embodiment.

REFERENCE SIGNS LIST

(48) 10 automatic analyzer 11 reagent container 12 reagent disk 13 reaction disk 14 reagent pipetting device 15 specimen pipetting device 16 conveyance line 17 cleaning tank 18 rack 19 specimen container 20 reaction cell 51 to 54 optical path 141 nozzle 142, 143 arm 144 shaft 151 nozzle 171 cleaning hole 200 imaging device 211 first mirror 212 second mirror 220 camera 221 image sensor 222 lens

Claims

1. An imaging device installed in a pipetting device including a nozzle configured to aspirate and dispense liquid and an arm configured to hold the nozzle and to move the nozzle by a rotation

operation, the imaging device comprising: a camera that includes an image sensor and a lens; and a pair of mirrors that includes a first mirror having a first reflective surface and a second mirror having a second reflective surface, wherein the pair of mirrors is arranged such that the first reflective surface and the second reflective surface face each other, and the pair of mirrors is arranged one by one between a tip of the nozzle and a bottom surface of the arm, on both sides across a plane that includes a rotation axis of the arm and a central axis of the nozzle, and the first mirror is disposed closer to the arm than the second mirror, with the first reflective surface facing toward the tip of the nozzle, and the second reflective surface facing toward the camera.

2. The imaging device according to claim 1, wherein the second mirror is disposed at a position closer to the plane than the first mirror.
3. The imaging device according to claim 2, wherein the second mirror is smaller than the first mirror.
4. The imaging device according to claim 1, wherein the imaging device is configured to be detachable from the pipetting device.
5. The imaging device according to claim 2, wherein an optical axis of the camera is disposed out of the plane including the rotation axis of the arm and the central axis of the nozzle.
6. The imaging device according to claim 1, wherein the imaging device is attached to the arm such that the camera is located near a base end portion of the nozzle.
7. The imaging device according to claim 1, wherein the lens is disposed such that light from a tip portion of the nozzle viewed from a first viewpoint is directly incident, and the pair of mirrors is arranged such that light from the tip portion of the nozzle at a second viewpoint different from the first viewpoint is reflected by the first mirror, then reflected by the second mirror, and incident on the lens.
8. The imaging device according to claim 7, wherein a direction of the first viewpoint is orthogonal to a direction of the second viewpoint.
9. The imaging device according to claim 7, further comprising a control device configured to process an image captured by the image sensor, wherein the control device is configured to calculate a deviation amount between the nozzle and a target stop position of the nozzle, which are projected in a region corresponding to the first viewpoint in the image, and the control device is configured to calculate a deviation amount between the nozzle and the target stop position, which are projected in a region corresponding to the second viewpoint in the image.
